

Accounting Business and Society

–Statement of Cash flows

WEEK 7

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Key concepts

- 01 Purpose & Format of Cash-flows statement
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- 04 Preparing for Investing Activities
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External Reports

Income
Statement

Balance Sheet

Statement of
Cash Flows

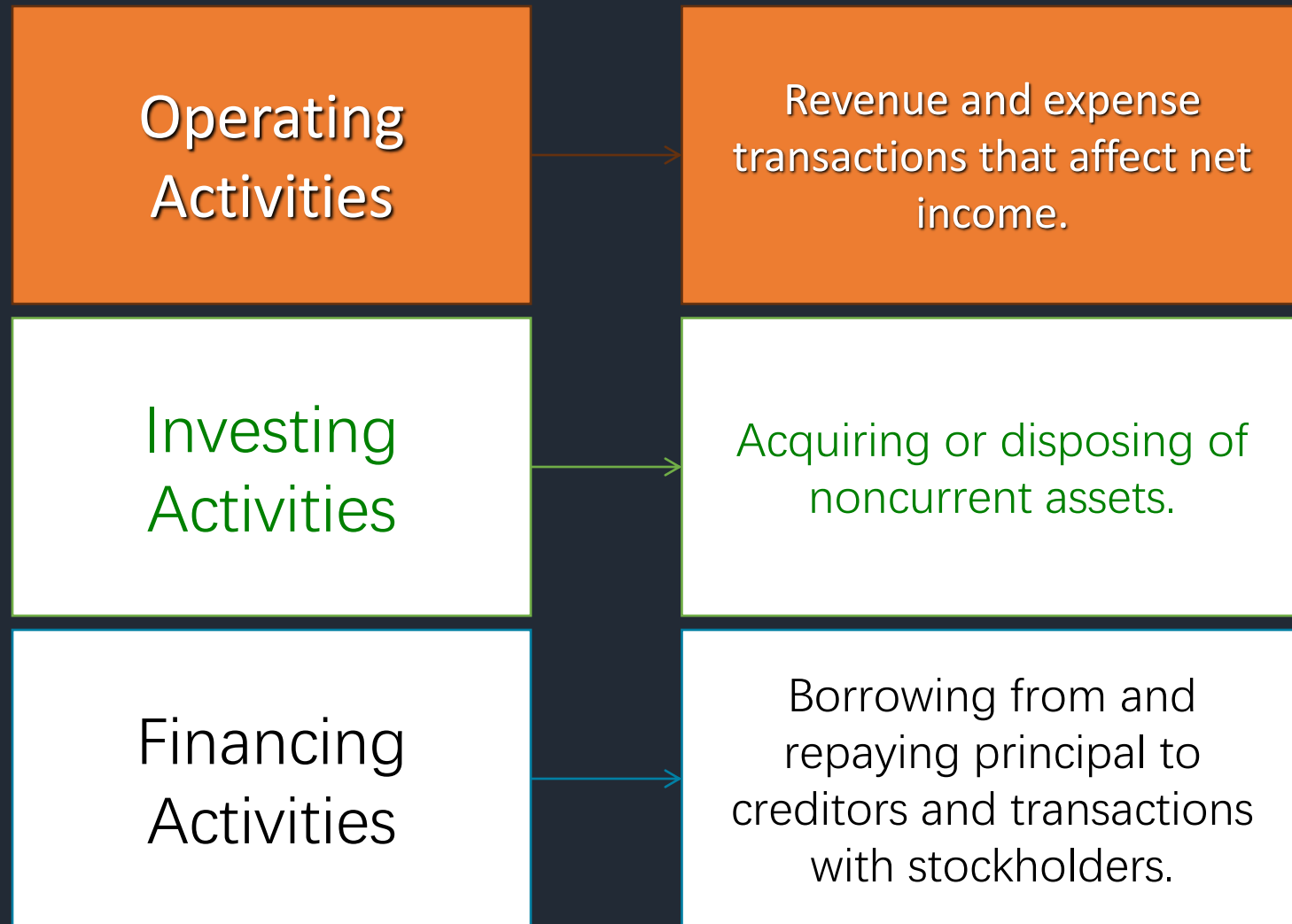
The statement of cash flows highlights the major activities that impact cash flows and hence, affect the overall cash balance.

1# Purpose of the Statement of Cash Flows

The statement of cash flows helps answer a variety of questions such as:

1. Are we generating sufficient cash flows from its ongoing operations to remain viable?
2. Can we pay our debts?
3. Can we pay our usual dividends?
4. Why do net income and net cash flow differ?
5. To what extent will we have to borrow money in order to make needed investments?

1# Format of Cash flow statement



1# Format of Cash flow statement

	Cash Inflow	Cash Outflow
Operating activities		
Collecting cash from customers	✓	
Paying suppliers for inventory purchases		✓
Paying bills to insurers, utility providers, etc		✓
Paying wages and salaries to employees		✓
Paying taxes to governmental bodies		✓
Paying interest to lenders		✓
Investing activities		
Buying property, plant, and equipment		✓
Selling property, plant, and equipment	✓	
Buying stocks and bonds as a long-term investment		✓
Selling stocks and bonds held for long-term investment	✓	
Lending money to another entity		✓
Collecting the principal on a loan to another entity	✓	
Financing activities		
Borrowing money from a creditor	✓	
Repaying the principal amount of a debt		✓
Collecting cash from the sale of common stock	✓	
Paying cash to repurchase your own common stock		✓
Paying a dividend to stockholders		✓

2# Process of preparing Cash-flows statement : A Fundamental Principle

$$\Delta \text{ Cash Balance} = \Delta \text{ Noncash Balance Sheet Accounts}$$

This principle ensures that properly analyzing the changes in all noncash balance sheet accounts always quantifies the cash inflows and outflows that explain the change in the cash balance.

A Review of Basic Equations

Basic Equation for Asset Accounts

Beginning balance + Debits – Credits = Ending balance

Basic Equation for Contra-Asset, Liability, and Stockholders' Equity Accounts

Beginning balance – Debits + Credits = Ending balance

Statement of Cash Flows: Key Concepts

The term ***cash*** on the statement of cash flows refers broadly to both currency and cash equivalents.

Currency and Bank Accounts = Cash

Treasury Bills, Commercial Paper, and
Money Market Funds = Cash Equivalents

Operating Activities – Direct or Indirect Method?

Direct Method

Reconstructs the income statement on a cash basis from top to bottom

Indirect Method

Accrual net income is adjusted to a cash basis; Used by 99%

Both methods result in the exact same amount of cash provided by operating activities.

3# Direct & Indirect method to prepare for OCF

Direct Method for Operating Cash ?...

- ☐ Report the **cash receipts** and **cash payments** from operating activities.
- ☐ Common items are shown in the template
- ☐ Using a set of formula to calculate

Cash Item	Formula to calculate
Receipts from customers	$\text{Sales} - \Delta \text{Account Receivables}$
Payment for Inventory	$\text{COGS} + \Delta \text{Inventory} - \Delta \text{Account Payable}$
Payment for Operating Expenses	$\text{Cash expenses} + \Delta \text{prepaid expenses} - \Delta \text{Payables}$
Dividend or Interest received	Obtained directly from Income Statement
Payment for Interest expense	$\text{Interest expense} - \Delta \text{Interest Payables}$
Payment for Tax expense	$\text{Income Tax expense} - \Delta \text{Income Tax Payables}$

Cash Flows from Operating Activities	
Receipts from customers	
Payments to suppliers	
Payments for expenses	
Dividends received	
Interest paid	
Income tax paid	
Net cash operating activities	XX

Important note:
 Δ = This year balance - Last year balance

3# Direct & Indirect method to prepare for OCF

Example ...

Financial Data
Linh's Accounting Business
for 2021-2022

	2022
	000's
Sales	2,511
Cost of sales	1,385
Gross profit	1,126
Other income	
Dividends received	25
Gain on sale of machinery	7
Total operating income	1,158
Operating Expenses	
Depreciation	68
Interest	21
Selling expenses	205
Administrative expenses	288
Total operating expenses	582
Profit before tax	576
Tax for the period	164
Profit for the period	412

	2022	2021
	\$000's	
Current Assets		
Cash and cash equivalents	382	342
Receivables	295	314
Inventory	540	485
Prepaid expenses	36	42
Total Current Assets	1,253	1,183
Non-current Assets		
Investments	162	108
Property, Plant and Equipment	468	460
Intangibles	140	140
Total Non-current Assets	770	708
Total Assets	2,023	1,891
Current Liabilities		
Trade Payables	294	286
Expenses payable	3	4
Income Taxes Payable	164	158
Total Current Liabilities	461	448
Non-current Liabilities		
Long Term Debt	228	257
Total Non-current Liabilities	228	257
Total Liabilities	689	705
Net Assets	1,334	1,186
Equity		
Share Capital	940	825
Retained earnings	394	361
Total Equity	1,334	1,186

Receipts from customers

$$= \text{Sales} - \Delta \text{Account Receivables}$$

$$= \$2511 - (\$295 - \$314) = \$2530$$

Payment for Inventory

$$= \text{COGS} + \Delta \text{Inventory} - \Delta \text{Account Payable}$$

$$= \$1385 + (\$540 - \$485) - (\$294 - \$286) = \$1432$$

3# Indirect method to prepare for OCF

Indirect Method for Operating Cash ?...

- ☐ Report cash flows from Operating activities by adjusting net income from **accrual basis** to **cash basis**
- ☐ **Adjustments** are made for :
 - Non-cash revenues (gain from sales)
 - Non-cash expenses (depreciation, losses)
 - Changes in Current Assets and Current Liabilities

Rules to make adjustments:

If CA		put -	▲
If CA		put +	▲
If CL		put +	▲
If CL		put -	▲

Formula to calculate

Net Income

Add : Depreciation expense (Loss from NCAs sold)

Less : Gain from Sales of NCAs

Adjustments for :

Current Assets (except Cash)

Current Liabilities

Net Cash flows from Operating Activities

The Indirect Method – A Three-Step Process

Step 1

**Add depreciation
charges to net
income.**



Step 2

**Analyze net changes
in noncash balance
sheet accounts.**



Step 3

**Adjust for gains and
losses.**

Step 1: Add Depreciation Charges – Part 1

Accumulated depreciation is a noncash balance sheet account and we must adjust net income for all of the changes in the noncash balance sheet accounts that have occurred during the period.

Basic Equation for Contra-Asset, Liability, and Stockholders' Equity Accounts

Beginning balance – Debits + Credits = Ending balance

Step 1: Add Depreciation Charges, Part 2

Account Activity for Accumulated Depreciation	
Beginning balance	\$300
Ending balance	\$500
Accumulated depreciation of equipment sold	\$70

Basic Equation for Contra-Asset, Liability, and Stockholders' Equity Accounts

Beginning balance – Debits + Credits = Ending balance

$$\$300 - \$70 + \text{Credits} = \$500$$

$$\text{Credits} = \$500 - \$300 + \$70$$

$$\text{Credits} = \$270$$

Step 2: Analyze Net Changes in Noncash Balance Sheet Accounts

	Increase in Account Balance	Decrease in Account Balance
Current Assets		
Accounts receivable	Subtract	Add
Inventory	Subtract	Add
Prepaid expenses	Subtract	Add
Current Liabilities		
Accounts payable	Add	Subtract
Accrued liabilities	Add	Subtract
Income taxes payable	Add	Subtract

Step 3: Adjust for Gains and Losses

Under U.S. GAAP and IFRS rules, gains and losses must be included in the **investing activities** section of the statement of cash flows.

Gains and losses must be removed from net income in the operating activities section before they can be shown in the investing activities section:

Subtract Gains
Add Losses

4# Preparing for ICF

U.S. GAAP and IFRS require that the investing and financing sections of the statement of cash flows disclose gross cash flows.

	Increase in Account Balance	Decrease in Account Balance
Noncurrent Assets (Investing activities)		
Property, plant, and equipment	Subtract	Add
Long-term investments	Subtract	Add
Loans to other entities	Subtract	Add
Liabilities and Stockholders' Equity (Financing activities)		
Bonds payable	Add	Subtract
Common stock	Add	Subtract
Retained earnings	*	*
*Requires further analysis to quantify cash dividends paid.		

Property, Plant, and Equipment

Account Activity for Property, Plant, and Equipment			
Beginning balance	\$1,000	Original cost of equipment sold	\$100
Ending balance	\$1,800	Accumulated depreciation of equipment sold	\$70
Cash proceeds from sale of equipment	\$40	Gain on the sale of equipment (included in net income)	\$10

Basic Equation for Asset Accounts

Beginning balance + Debits – Credits = Ending balance

$$\$1,000 + \text{Debits} - \$100 = \$1,800$$

$$\text{Debits} = \$1,800 - \$1,000 + \$100$$

$$\text{Debits} = \$900 \text{ (cash outflow)}$$

Report \$40
cash inflow.

Report \$900
cash outflow.

5# Preparing for FCF

U.S. GAAP and IFRS require that the investing and financing sections of the statement of cash flows disclose gross cash flows.

	Increase in Account Balance	Decrease in Account Balance
Noncurrent Assets (Investing activities)		
Property, plant, and equipment	Subtract	Add
Long-term investments	Subtract	Add
Loans to other entities	Subtract	Add
Liabilities and Stockholders' Equity (Financing activities)		
Bonds payable	Add	Subtract
Common stock	Add	Subtract
Retained earnings	*	*
*Requires further analysis to quantify cash dividends paid.		

Retained Earnings

Account Activity for Retained Earnings	
Beginning balance	\$2,000
Ending balance	\$3,000
Net income	\$1,200

Basic Equation for Contra-Asset, Liability, and Stockholders' Equity Accounts

Beginning balance – Debits + Credits = Ending balance

$$\$2,000 - \text{Debits} + \$1,200 = \$3,000$$

$$\$3,200 = \$3,000 + \text{Debits}$$

$$\text{Debits} = \$200 \text{ (cash outflow)}$$

Report \$1,200 net income
in Operating Activities.

Report \$200 dividends paid
in Financing Activities.

Summary of Key Concepts – Part 1

Key Concept #1		Key Concept #2	
The statement of cash flows is divided into three sections:		U.S. GAAP and IFRS allow two methods for preparing the operating activities section of the statement of cash flows:	
Operating activities		Direct Method (Appendix 14A)	
Net cash provided by (used in) operating activities	\$xx	Cash receipts from customers	\$ xx
Investing activities		Cash paid for inventory purchases	(xx)
Net cash provided by (used in) investing activities	xx	Cash paid for selling and administrative expenses	(xx)
Financing activities		Cash paid for income taxes	<u>(xx)</u>
Net cash provided by (used in) financing activities	<u>xx</u>	Net cash provided by (used in) operating activities	\$ xx
		Indirect Method	
Net increase/decrease in cash and cash equivalents	xx	Net income	\$ xx
Beginning cash and cash equivalents	<u>xx</u>	Various adjustments (+/–)	<u>xx</u>
Ending cash and cash equivalents	<u>\$xx</u>	Net cash provided by (used in) operating activities	<u>\$ xx</u>

Summary of Key Concepts – Part 2

Key Concept #3		Key Concept #4	
Computing the net cash provided by operating activities using the indirect method is a three step process:		The investing and financing sections of the statement of cash flows must report gross cash flows:	
Step 1	Operating activities	Net cash provided by (used in) operating activities	\$xx
	Net income		
	Adjustments to convert net income to a cash basis:	Investing activities	
		Purchase of property, plant, and equipment	(xx)
		Sale of property, plant, and equipment	xx
Step 2	Add: Depreciation	Purchase of long-term investments	(xx)
		Sale of long-term investments	<u>xx</u>
		Net cash provided by (used in) investing activities	(xx)
	Analyze net changes in noncash balance sheet accounts:	Financing activities	
	Increase in current asset accounts	Issuance of bonds payable	xx
Step 3	Decrease in current asset accounts	Repaying principal on bonds payable	(xx)
	Increase in current liability accounts	Issuance of common stock	xx
	Decrease in current liability accounts	Purchase own shares of common stock	(xx)
	Adjust for gains/losses:	Paying a dividend	<u>(xx)</u>
	Gain on sale	Net cash provided by (used in) financing activities	<u>xx</u>
	Loss on sale	Net increase/decrease in cash and cash equivalents	xx
	Net cash provided by (used in) operating activities	Beginning cash and cash equivalents	<u>xx</u>
		Ending cash and cash equivalents	<u>\$xx</u>

Example

An Example of a Statement of Cash Flows:

Apparel, Inc. Financial Statements – Part 1

Apparel, Inc. Income Statement (dollars in millions)	
Sales	\$3,638
Cost of goods sold	<u>2,469</u>
Gross margin	1,169
Selling and administrative expenses	<u>941</u>
Net operating income	228
Nonoperating items: Gain on sale of store	<u>3</u>
Income before taxes	231
Income taxes	<u>91</u>
Net income	<u><u>\$ 140</u></u>

Apparel, Inc. Financial Statements – Part 2

Apparel, Inc. Comparative Balance Sheet (dollars in millions)

	Ending Balance	Beginning Balance	Change
Assets			
Current assets:			
Cash and cash equivalents	\$ 91	\$ 29	+62
Accounts receivable	637	654	-17
Inventory	586	537	+49
Total current assets	<u>1,314</u>	<u>1,220</u>	
Property, plant, and equipment	1,517	1,394	+123
Less accumulated depreciation	654	561	+93
Net property, plant, and equipment	<u>863</u>	<u>833</u>	
Total assets	<u>\$2,177</u>	<u>\$2,053</u>	
Liabilities and Stockholders' Equity			
Current liabilities:			
Accounts payable	\$ 264	\$ 220	+44
Accrued liabilities	193	190	+3
Income taxes payable	75	71	+4
Total current liabilities	<u>532</u>	<u>481</u>	
Bonds payable	<u>479</u>	<u>520</u>	-41
Total liabilities	<u>1,011</u>	<u>1,001</u>	
Stockholders' equity:			
Common stock	157	155	+2
Retained earnings	<u>1,009</u>	<u>897</u>	+112
Total stockholders' equity	<u>1,166</u>	<u>1,052</u>	
Total liabilities and stockholders' equity	<u>\$2,177</u>	<u>\$2,053</u>	

An Example of a Statement of Cash Flows

In addition to the financial statements provided, assume the following:

1. The company sold a store that had an original cost of \$15 million and accumulated depreciation of \$10 million. The cash proceeds from the sale were \$8 million. The gain on the sale was \$3 million.
2. The company did not issue any new bonds during the year.
3. The company did not repurchase any of its own common stock during the year.
4. The company paid a cash dividend during the year.

Operating Activities – Step 1

The first step in computing Apparel's net cash provided by operating activities is to add depreciation to net income.

Basic Equation for Contra-Asset, Liability, and Stockholders' Equity Accounts

Beginning balance – Debits + Credits = Ending balance

$$\$561\text{M} - \$10\text{M} + \text{Credits} = \$654\text{M}$$

$$\text{Credits} = \$654\text{M} - \$561\text{M} + \$10\text{M}$$

$$\text{Credits} = \$103\text{M}$$

Operating Activities – Step 2

The second step in computing Apparel's net cash provided by operating activities is to analyze net changes in noncash balance sheet accounts that impact net income.

	Increase in Account Balance	Decrease in Account Balance
General Guidelines from Exhibit 14-2		
Current Assets:		
Accounts receivable	Subtract	Add
Inventory	Subtract	Add
Current Liabilities:		
Accounts payable	Add	Subtract
Accrued liabilities	Add	Subtract
Income taxes payable	Add	Subtract
	Increase in Account Balance	Decrease in Account Balance
Apparel's Account Analysis		
Current Assets:		
Accounts receivable		→ +17
Inventory	→ -49	
Current Liabilities:		
Accounts payable	+44	
Accrued liabilities	+3	
Income taxes payable	+4	

Operating Activities – Step 3

The third step in computing Apparel's net cash provided by operating activities is to adjust for gains and losses included in net income.

Apparel, Inc. Income Statement (dollars in millions)

Sales	\$3,638
Cost of goods sold	<u>2,469</u>
Gross margin	1,169
Selling and administrative expenses	<u>941</u>
Net operating income	228
Nonoperating items: Gain on sale of store	<u>3</u>
Income before taxes	231
Income taxes	<u>91</u>
Net income	<u><u>\$ 140</u></u>

Operating Activities

Apparel, Inc.
(dollars in millions)

Operating Activities

	Net income		\$140
	Adjustments to convert net income to a cash basis:		
Step 1 →	Depreciation	103	
	Decrease in accounts receivable	17	
	Increase in inventory	(49)	
Step 2 →	Increase in accounts payable	44	
	Increase in accrued liabilities	3	
	Increase in income taxes payable	4	
Step 3 →	Gain on sale of store	(3)	119
	Net cash provided by (used in) operating activities .		<u>\$259</u>

Investing Activities

Basic Equation for Asset Accounts

Beginning balance + Debits – Credits = Ending balance

$\$1,394\text{M} + \text{Debits} - \$15\text{M} = \$1,517\text{M}$

$\text{Debits} = \$1,517\text{M} - \$1,394\text{M} + \$15\text{M}$

$\text{Debits} = \$138\text{M}(\text{cash outflow})$

Report
\$8 million cash
inflow.

Report
\$138 million
cash outflow.

Financing Activities

	Increase in Account Balance	Decrease in Account Balance
General Guidelines from Exhibit 14-3		
Liabilities and Stockholders' Equity:		
Bonds payable	Add	Subtract
Common stock	Add	Subtract
	Increase in Account Balance	Decrease in Account Balance
Apparel's Account Analysis		
Liabilities and Stockholders' Equity:		
Bonds payable		
Common stock	→ +2	→ -41

Basic Equation for Contra-Asset, Liability, and Stockholders' Equity Accounts

Beginning balance – Debits + Credits = Ending balance

$$\$897\text{M} - \text{Debits} + \$140\text{M} = \$1,009\text{M}$$

$$\$1,037\text{M} = \$1,009\text{M} + \text{Debits}$$

$$\text{Debits} = \$28\text{M (cash outflow)}$$

Apparel, Inc. – Statement of Cash Flows

Apparel, Inc.
Statement of Cash Flows—Indirect Method
(dollars in millions)

Operating Activities

Net income.....		\$ 140
Adjustments to convert net income to a cash basis:		
Depreciation.....	103	
Decrease in accounts receivable.....	17	
Increase in inventory.....	(49)	
Increase in accounts payable.....	44	
Increase in accrued liabilities.....	3	
Increase in income taxes payable.....	4	
Gain on sale of store.....	(3)	119
Net cash provided by (used in) operating activities.....		<u>259</u>

Investing Activities

Additions to property, plant, and equipment.....	(138)	
Proceeds from sale of store.....	8	
Net cash provided by (used in) investing activities.....		<u>(130)</u>

Financing Activities

Retirement of bonds payable.....	(41)	
Issuance of common stock.....	2	
Cash dividends paid.....	(28)	
Net cash provided by (used in) financing activities.....		<u>(67)</u>
Net increase in cash and cash equivalents.....		62
Beginning cash and cash equivalents.....		<u>29</u>
Ending cash and cash equivalents.....		<u>\$ 91</u>

Seeing the Big Picture

Cash					
Net income	(1)	140	49	(4)	Increase in inventory
Depreciation	(2)	103	3	(12)	Gain on sale of store
Decrease in accounts receivable	(3)	17			
Increase in accounts payable	(5)	44			
Increase in accrued liabilities	(6)	3			
Increase in income taxes payable	(7)	4			
Net cash provided by operating activities		259			
Proceeds from sale of store	(12)	8	138	(8)	Additions to property, plant, and equipment
Increase in common stock	(11)	2	41	(9)	Decrease in bonds payable
			28	(10)	Cash dividends paid
Net increase in cash and cash equivalents		62			

Accounts Receivable			Inventory			Property, Plant, and Equipment			Accumulated Depreciation		
Bal.	654		Bal.	537		Bal.	1,394			561	Bal.
	17	(3)	(4)	49		(8)	138	15	(12)	103	(2)
Bal.	637		Bal.	586		Bal.	1,517			654	Bal.

Accounts Payable			Accrued Liabilities			Income Taxes Payable		
	220	Bal.		190	Bal.		71	Bal.
	44	(5)		3	(6)		4	(7)
	264	Bal.		193	Bal.		75	Bal.

Bonds Payable			Common Stock			Retained Earnings		
	520	Bal.		155	Bal.		897	Bal.
(9)	41			2	(11)	(10)	140	(1)
	479	Bal.		157	Bal.		1,009	Bal.

Extention

Compute free cash flow.

Free Cash Flows – Part 1

Free cash flow measures a company's ability to fund its capital expenditures and dividends from its net cash provided by operating activities.

$$\text{Free Cash Flow} = \text{Net Cash Provided by Operating Activities} - \text{Capital Expenditures} - \text{Dividends}$$

Free Cash Flows – Part 2

Free cash flow measures a company's ability to fund its capital expenditures and dividends from its net cash provided by operating activities.

Free Cash Flow	=	Net Cash Provided by Operating Activities	-	Capital Expenditures	-	Dividends
\$ 93	=	\$ 259	-	\$ 138	-	\$ 28

Earnings Quality

Managers generally perceive that earnings are of higher quality when the earnings:

1. are not unduly influenced by inflation,
2. are computed using conservative accounting principles and estimates, and
3. are correlated with net cash provided by operating activities.

Assessment

Tutorial portfolio - Online quizzes:

This quiz is based on the week 7 learning materials. You may complete this anytime during week 7, you may use your study notes.

You have TWO attempts to complete this quiz.

Due time for your weekly quiz is on Canvas, please check it by yourself.